

COMPUTATIONAL GENOMICS FINAL PROJECT

Who left this DNA?

A mystery Nanopore fecal sample, six possible host animals, one microbial co-star, and a small pile of evidence.

Group D

Host origin, microbiome, assembly, and validation from curated project outputs.

THE BRIEF

What the assignment actually asked us to do

Technical part

Clean and summarize the Nanopore reads. Explain why the data are usable.

Bioinformatics part

Find the host animal, classify microbes, assemble one abundant species, and validate it.

Biological part

Turn the outputs into a story: what animal, what microbe, what functional region, and how confident we are.

6

candidate host genomes

300k

reads used for the first host screen

1

dominant microbial species

1

reproducible notebook pipeline

Six biology words translated for a CS room

Read

One DNA fragment reported by the sequencer. In CS terms: one noisy string sampled from a much larger text.

Reference genome

A known genome used as a coordinate system. We ask: where would this read fit best?

Alignment

Approximate string matching between a read and a reference genome.

MAPQ

A confidence score from the aligner. High MAPQ means the read has a clearer best location.

Locus

A genomic address or interval, written like reference:start-end.

Gene annotation

A map that labels some intervals as genes or regulatory features. Without it, coordinates are just coordinates.

ANALYSIS MAP

From messy reads to a biological story

1

Check the reads

Are they long enough and clean enough to trust?

2

Find the host

Align reads to the six candidate genomes.

3

Pick a region

Use gene annotations to find a readable functional-region example.

4

Classify microbes

Ask what else appears in the fecal DNA.

5

Assemble and validate

Reconstruct the dominant microbe and compare it to a reference.

The goal was not to make a fancy tool zoo. It was to make each step answer the next practical question.

What the software did, before we throw names around

MINIMAP2

Long-read aligner

Finds where noisy Nanopore reads best match a reference genome.

KRAKEN2

Taxonomy classifier

Breaks reads into short k-mers and votes for likely microbes using a database.

FLYE

Genome assembler

Turns many overlapping long reads into longer reconstructed chunks called contigs.

DNADIFF

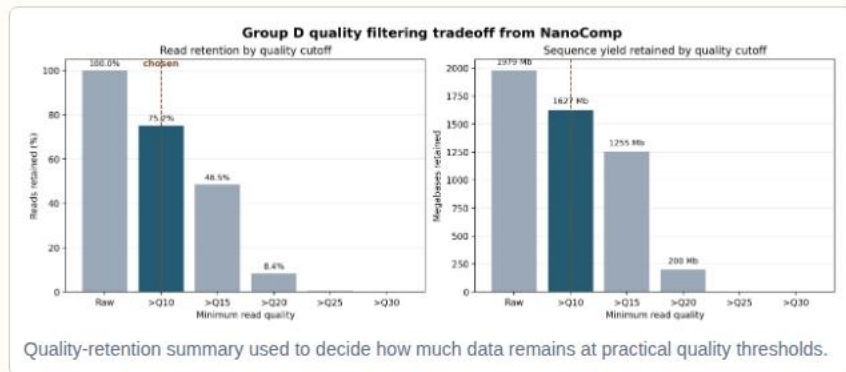
Assembly checker

Compares our assembled sequence to a known reference and reports identity/coverage.

The names are less important than the questions: align, classify, assemble, compare.

QUALITY CONTROL

First question: are the reads usable?



Answer:

Yes. The reads are noisy, as Nanopore reads usually are, but still usable for long-read alignment and microbial screening.

Why these reports?

MinKNOW describes the sequencing run itself. NanoComp and NanoPlot summarize the reads after the fact, which is better for reproducible QC in the notebook. FastQC is common for short reads, but less informative for Nanopore length distributions.

Before interpreting biology, we looked at the run reports



These browser reports are the quick visual sanity check before we trust downstream analysis.

SUSPECT LINEUP

Six animals walked into the alignment step



House mouse

Good cameo, weak evidence.



Kangaroo

Memorable, but not our sample.



Domestic cat

Quietly sitting on the strongest signal.



Norway rat

Similar enough to test, not enough to win.



Golden jackal

Second place in some counts.



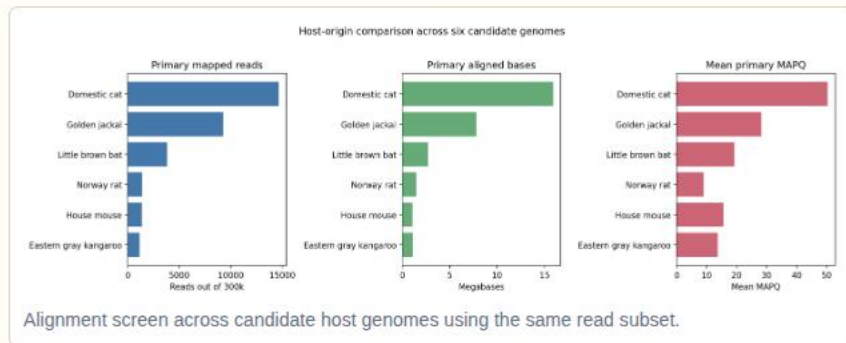
Little brown bat

Useful negative control energy.

The mouse got a slide. The cat got the evidence.

FIRST TRY

Let the reads vote: which genome do they align to?



Domestic cat had the strongest host signal.

It had the most primary mapped reads and the most aligned bases among the candidate genomes.

Why minimap2?

It is fast, widely used, and tuned for long noisy reads. BWA-MEM is excellent for many Illumina-style tasks, but minimap2 is the practical default for Nanopore host screening. BLAST would be too slow at this scale.

SECOND TRY

Not just more reads. Better reads.

Candidate	Primary mapped	Aligned bases	Mean MAPQ
1 Domestic cat	14,669	15.9 Mbp	50.3
2 Golden jackal	9,251	7.8 Mbp	28.3
3 Little brown bat	3,818	2.7 Mbp	19.2
Mouse / rat / kangaroo	1,155-1,393	1.1-1.5 Mbp	9.0-15.6

MAPQ is the confidence check.

A higher MAPQ means the aligner is more confident that a read belongs at that location.

Cat wins twice.

More mapped signal and much higher confidence than the nearest alternative.

HOST REVEAL

The sample origin is domestic cat

The candidate lineup was fun. The alignment scorecard made the call.

Top mapped reads

Top aligned bases

Top MAPQ



What does "host locus" mean here?

A locus is just an address in the host genome.

After we decided the host was cat, we asked: which annotated cat genome intervals overlap our cat-aligned reads?

"Functional" means the address has a known label.

The annotation file labels some intervals as genes. Those labels let us say something biological instead of only reporting coordinates.

1

Cat-aligned reads

Start only after the host call.

2

Cat annotation file

A table of gene coordinates from NCBI.

3

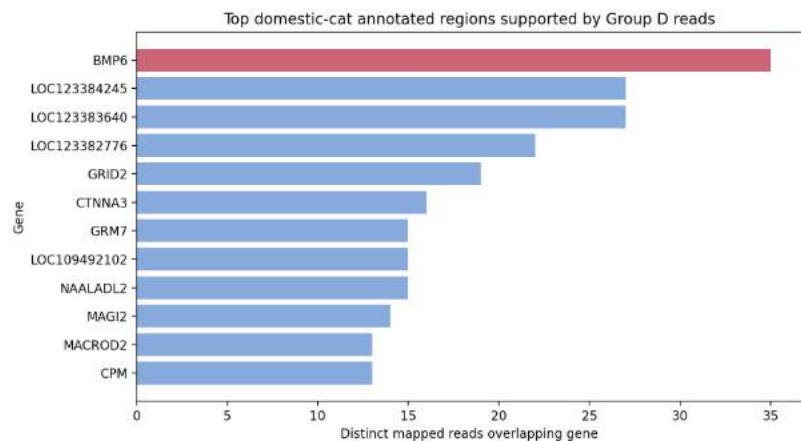
Overlap count

Which labeled region received the most read support?

So "host locus" is not a new organism and not a phenotype claim. It is a genomic address inside the cat reference.

WHY BMP6?

BMP6 was our best annotated-region example



BMP6 had the highest read count among annotated cat gene overlaps in our summary.

Why it was chosen:

It was the top labeled gene region in the overlap table: 35 reads overlapped the BMP6 interval.

What BMP6 is:

BMP6 is a mammalian gene in a signaling family often discussed in development, bone/cartilage biology, and iron regulation.

What we are not saying:

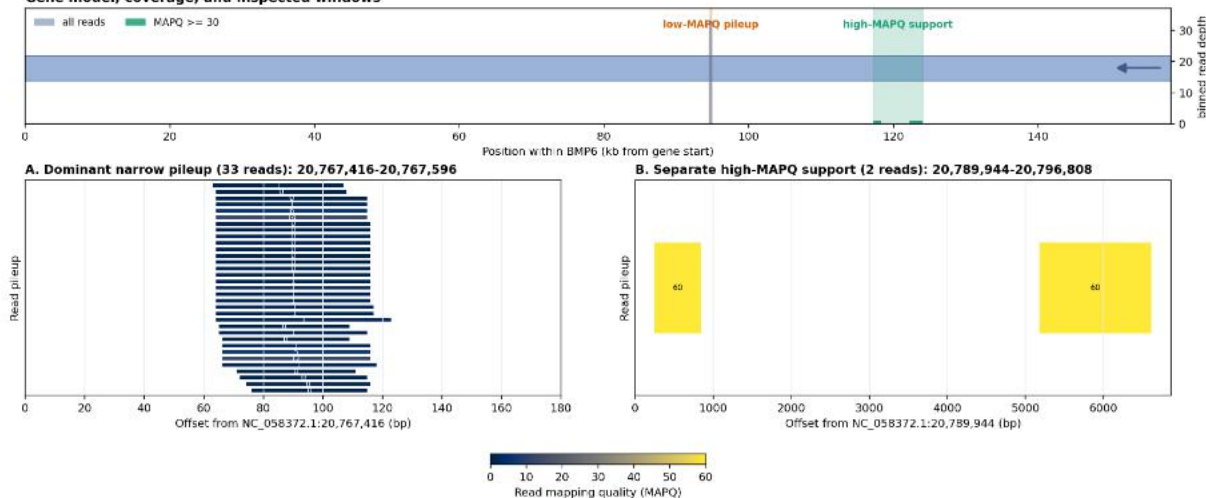
We are not saying this cat had a BMP6 mutation, phenotype, or special biology. We only found that this annotated region was represented in the host reads.

A genome-browser view makes the BMP6 claim less abstract

BMP6 host-region evidence: genome-browser-style view

NC_058372.1:20,672,770-20,831,131 | 158.4 kb gene | 35 overlapping reads | 2 reads with MAPQ ≥ 30

Gene model, coverage, and inspected windows



IGV-style BMP6 locus panel. The top track is the genomic address, the middle shows read pileup, and the annotation track marks the gene interval.

Why an IGV-style panel?

Summary tables can hide awkward alignments. A genome-browser view lets us see reads sitting on a coordinate system, like a debugger for genome results.

Read it carefully:

The signal is concentrated and should be treated as supporting evidence, not as a functional discovery.

A fecal sample rarely travels alone

After host screening, the next question was: who else is in there?

For that, we used taxonomic classification rather than host-style whole-genome alignment.

Why Kraken2?

Kraken2 is fast and works well when the question is "what taxa are represented here?" Centrifuge is another lightweight classifier, MetaPhlAn is strong for marker-gene profiles, and KrakenUniq adds extra support information. Kraken2 was the best fit for a quick, reproducible species screen.

Plain version:

Host alignment asks "which animal genome?" Kraken2 asks "which microbial names do these reads most resemble?"

How a read becomes a microbial name

1

Cut into k-mers

Each read is split into many short DNA words.

2

Look up words

The database links DNA words to organisms.

3

Vote carefully

If enough words point to the same taxon, Kraken2 reports that label.

Why this fits the task:

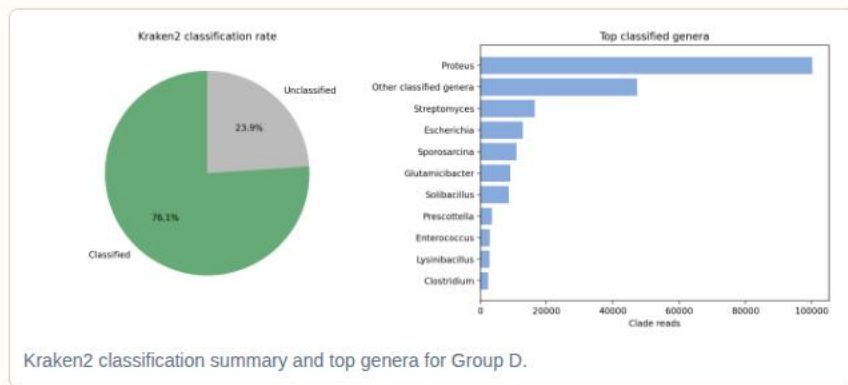
The assignment asks for microbiome composition, diversity, and top taxa. Classification is exactly that question.

Why we still keep caveats:

The answer depends on the database. Closely related organisms can share many DNA words.

MICROBIAL SCORECARD

Proteus dominates the genus-level profile



35.11%

Proteus at genus level

5.77%

Streptomyces

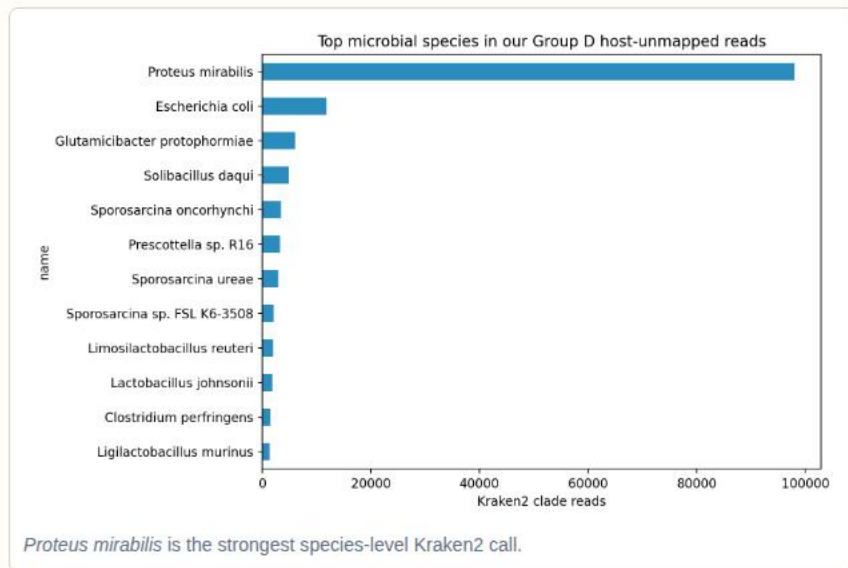
4.49%

Escherichia

3.84%

Sporosarcina

The dominant species is *Proteus mirabilis*



Top species call:

Proteus mirabilis, 34.35% of reads in the Kraken2 report.

The biological story so far:

A domestic cat fecal sample with a strong *Proteus mirabilis* microbial signal.

What does it mean to "assemble a genome"?

1

Reads are pieces

Each read is a noisy fragment from the microbial genome.

2

Overlaps connect pieces

If two reads share sequence, they may come from neighboring positions.

3

Contigs are longer chunks

The assembler outputs reconstructed sequences, not necessarily one perfect chromosome.

What Flye did:

Flye used the long noisy reads to build those longer contigs. It is the reconstruction step, not the classifier and not the validator.

Why Flye?

Flye is built for noisy long reads and is commonly used for bacterial Nanopore assemblies. Canu is powerful but heavier, Raven is fast and lightweight, and Shasta is excellent in some long-read contexts. Flye was a balanced choice for this final-project dataset.

ASSEMBLY TARGET

Can we reconstruct the microbial co-star?

Why assemble *Proteus mirabilis*?

It had the strongest species-level signal, so it was the best candidate for a focused assembly attempt.

What we expected:

Because this is a mixed fecal sample, a fragmented assembly is normal. A perfect one-contig genome would be surprising.

3.86 Mbp

assembly size

24

contigs

577,987 bp

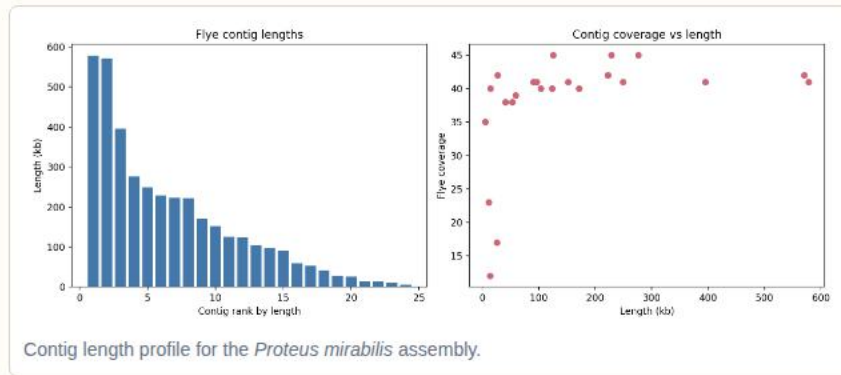
longest contig

249,377 bp

N50

ASSEMBLY RESULT

The assembly is fragmented, but biologically useful



What this means:

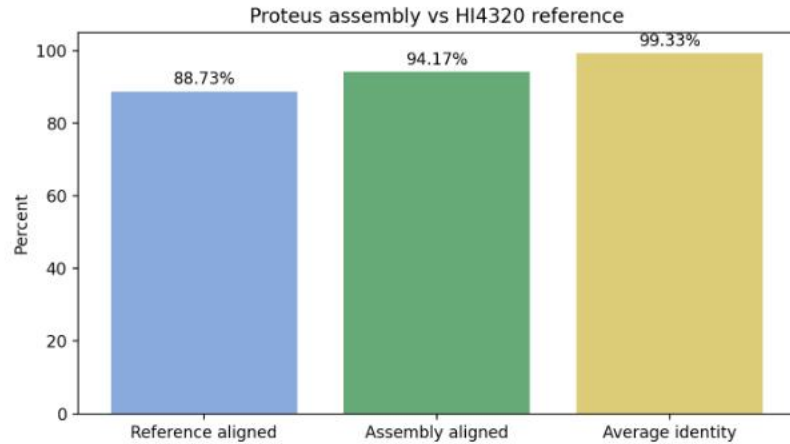
A perfect bacterial assembly would be closer to one complete chromosome. Here we got a partial but coherent assembly from a mixed fecal sample.

Good enough for:

Checking whether the dominant microbe can be reconstructed and compared against a reference.

VALIDATION

Does the assembly look like the reference?



dnadiff comparison between the assembly and a *Proteus mirabilis* reference.

Average identity: 99.33%

The assembled sequence aligns very closely to the reference where it aligns.

Coverage is not complete.

88.73% of the reference and 94.17% of the assembly aligned in the comparison.

Why dnadiff?

dnadiff is like a whole-genome diff tool. It aligns our assembly to a reference and reports how much matches and how similar the aligned parts are. QUAST is a strong alternative for broader assembly QC.

WHAT WE TRUST

Strong conclusions, careful caveats

We trust these results

- Host origin is domestic cat based on read count, aligned bases, and MAPQ.
- *Proteus mirabilis* is the dominant species-level microbial call.
- The Proteus assembly is close to a reference over aligned regions.

We are careful about these points

- BMP6 is a selected functional-region example from read overlap, not a phenotype claim.
- Kraken2 depends on the database and can misclassify reads from close relatives.
- The assembly is fragmented and from a mixed sample, so it is not a finished genome.

FINAL ANSWER

The short version

Host:

Domestic cat.

Functional region example:

BMP6 gene region, chosen because it had the strongest annotated host-read overlap.

Dominant microbe:

Proteus mirabilis.

Assembly:

3.86 Mbp, 24 contigs, N50 249,377 bp, 99.33% identity.

The cat did it.

Proteus mirabilis left the microbial fingerprints.

Full report

Curated GitHub Pages report

Notebook

Rendered reproducible analysis

PDF deck

Downloadable presentation